

components

3×3 conv

1×1 conv

BN

ReLU

pool / GAP

each operator has one job;
the chapter changes how they are composed

reusable block

VGG block

$[3 \times 3 \text{ conv} \rightarrow \text{BN} \rightarrow \text{ReLU}] \times 2 \rightarrow \text{pool}$

NiN block

$3 \times 3 \text{ conv} \rightarrow 1 \times 1 \rightarrow 1 \times 1$

residual block

$3 \times 3 \rightarrow 3 \times 3 \rightarrow +$

identity skip

architecture = stacked blocks + head

VGG block

$\times L$

flatten
dense head

NiN block

$\times L$

global average
pool + logits

residual block

$\times L$

global average
pool + logits

blocks preserve a bird's-eye contract:
spatial size, channel width, and route for information